MockTIA Virtual Labs

LAB 2.1.16 | SET UP A DESKTOP COMPUTER

Student Study Guide + Vocabulary + Review Quiz • Created by Mr. Beaird

Learning goal: Prepare a desktop workstation for a new employee. Connect desktop and monitor power, HDMI video, USB keyboard and mouse, and RJ-45 Ethernet. Boot the computer and check connectivity using Windows Command Prompt.

What you must be able to do

• Identify each cable and the port it belongs in.

• Connect everything safely before powering on the workstation.

• Explain the difference between AC power, HDMI video, USB input, and Ethernet networking.

• Boot Windows and verify a local network address and connectivity.

• Use **ipconfig**, **ipconfig /all**, and **ping** to confirm network operation.

Connection map

Device / cable	Connect to	Why
PC AC power cord	PC power input / outlet	Supplies electrical power to the PC
Monitor AC power cord	Monitor power input / outlet	Powers the monitor separately
HDMI cable	PC video output → monitor HDMI input	Carries the display signal
USB keyboard	PC USB port	Allows keyboard input
USB mouse	PC USB port	Allows pointer input
Ethernet (RJ-45)	PC Ethernet port → network jack/switch	Connects the PC to the LAN

Safe setup sequence

1. Place computer and monitor securely and identify all ports.

2. Connect monitor power and PC power without energizing the workstation yet.

3. Connect HDMI between the PC display output and monitor input.

4. Connect keyboard and mouse to USB ports.

5. Connect Ethernet to the PC network port and wall jack/switch.

6. Power on the monitor and computer; check for successful startup.

7. Open Command Prompt and verify addressing and connectivity.

8. Run the lab Check Answers function, enter your full name, and print the completion report.

Vocabulary reference

Know these terms before attempting the lab.

Desktop workstation — A computer and its connected devices used to perform work.

Peripheral — An external device connected to a computer, such as a keyboard or mouse.

HDMI — High-Definition Multimedia Interface; commonly carries digital video and audio.

USB — Universal Serial Bus; connects input devices and many other peripherals.

RJ-45 / 8P8C connector — Common modular plug used for twisted-pair Ethernet cabling.

Ethernet — Wired networking technology commonly used on local-area networks.

LAN — Local Area Network; devices connected within one site or network segment.

NIC — Network Interface Card/controller; the hardware connecting a computer to a network.

AC power — Alternating-current electricity supplied by the wall outlet.

POST — Power-On Self-Test; basic hardware checks during startup.

Boot — Process of starting the computer and loading the operating system.

IP address — Logical network address assigned to an interface.

Subnet mask — Defines the network and host portions of an IPv4 address.

Default gateway — Router address used to reach destinations outside the local network.

DNS — Domain Name System; translates host names into IP addresses.

Command Prompt — Windows command-line interface for diagnostics and administration.

ipconfig — Windows command displaying IP configuration details.

ping — Command used to test reachability to another network device using ICMP.

Troubleshooting — A methodical process of identifying and correcting a problem.

Verification — Confirming that the repair or setup meets the required objective.

Windows verification and troubleshooting

These commands are supported by the MockTIA desktop setup simulation.

`ipconfig` — Display basic IP address and subnet information.

`ipconfig /all` — Display detailed adapter addressing, gateway, and DNS information.

`ping 192.168.0.10` — In this simulation, check connectivity to the local server.

`ping 198.28.2.254` — In this simulation, check connectivity to the external/ISP test address.

`help` — See available simulated commands.

`cls` — Clear the simulated Command Prompt screen.

Symptom → what to check

PC will not turn on: Check PC AC power cable, outlet/power source, and the power button.

Monitor is dark: Check separate monitor AC power and power switch.

Monitor says “No Signal”: Check HDMI cable at both ends, the chosen input, and that the PC is running.

Keyboard or mouse does not respond: Check USB connection and the device/port.

No network connectivity: Check the Ethernet cable/port, then use ipconfig and ping.

Can ping local host but not remote target: Check gateway, routing, and upstream network availability.

Practice review (answer before looking at the key)

1. What cable connects a desktop PC to a monitor for digital video? _____

2. Why do the PC and monitor each need a power connection? _____

3. What does RJ-45 Ethernet connect the desktop to? _____

4. Which Windows command shows detailed IP settings? _____

5. What does a successful ping help demonstrate? _____

6. Why should you avoid forcing a connector into a port? _____

7. If a PC boots but the screen is black, what two connections would you check first? _____

8. Why is verifying functionality after setup important? _____

Answer key and completion checklist

1. HDMI.

2. The PC and monitor are separately powered devices.

3. A network wall jack or switch for LAN connectivity.

4. ipconfig /all.

5. That an IP target responds to an ICMP echo request (within the simulation, reachability).

6. The wrong connector can damage the connector or port.

7. Monitor power and HDMI connection/input (also confirm the PC is running).

8. It confirms the setup meets the work order and catches unresolved problems.

Student lab checklist

■ PC AC power connected

■ Monitor AC power connected

■ HDMI video cable connected

■ USB keyboard connected

■ USB mouse connected

■ Ethernet cable connected

■ Computer boots successfully

■ Local and external connectivity verified

■ Student full name entered and completion report saved as PDF

Submission: Complete the HTML simulation, use its actual assessment/print option, choose “Save as PDF,” and submit the resulting completion report in Canvas. This guide is for studying; it is not proof that the lab was completed.